a
B. subtilis growth (h−1)
CarveMe†
7 better / 0 worse / 0 equal
Glc
Glc+Mal
Fru
Glyc
Mal
Succ+Glu
Pyr
0.0 0.2 0.4 0.6
BiGG
3 better / 5 worse / 0 equal
Fru
Glyc
Glc+Mal
6pgc
Glc
Mal
Succ+Glu
Pyr
0.0 0.2 0.4 0.6
ModelSEED†
6 better / 0 worse / 0 equal
Glc+Mal
Mal
Glc
Fru
Succ+Glu
Glyc
0.0 0.2 0.4 0.6
VMH†
6 better / 0 worse / 0 equal
Glc+Mal
Mal
Glc
Fru
Glyc
Succ+Glu
0.0 0.2 0.4 0.6
b
E. coli exchange fluxes (mmol gDCW−1 h−1)
CarveMe
24 better / 0 worse / 0 equal
CO2 0.40
CO2 0.35
Glc 0.40
CO2 0.30
Glc 0.35
Glc 0.30
CO2 0.25
Glc 0.25
CO2 0.20
Glc 0.20
CO2 0.15
O2 0.40
Glc 0.15
O2 0.35
O2 0.30
O2 0.25
O2 0.20
O2 0.15
NH4+ 0.25
NH4+ 0.40
NH4+ 0.30
NH4+ 0.20
NH4+ 0.35
NH4+ 0.15
−2 0 2 4 6
BiGG
8 better / 10 worse / 6 equal
Glc 0.40
Glc 0.35
Glc 0.30
Glc 0.25
Glc 0.20
Glc 0.15
CO2 0.40
CO2 0.35
NH4+ 0.15
NH4+ 0.20
NH4+ 0.25
NH4+ 0.30
NH4+ 0.35
NH4+ 0.40
O2 0.40
CO2 0.15
O2 0.15
O2 0.35
CO2 0.30
CO2 0.20
O2 0.20
CO2 0.25
O2 0.25
O2 0.30
−2 0 2 4 6
ModelSEED
16 better / 0 worse / 8 equal
CO2 0.30
CO2 0.40
CO2 0.35
CO2 0.25
CO2 0.20
CO2 0.15
Glc 0.25
Glc 0.40
Glc 0.30
Glc 0.20
Glc 0.35
Glc 0.15
O2 0.25
O2 0.30
O2 0.35
O2 0.40
O2 0.15
O2 0.20
NH4+ 0.15
NH4+ 0.20
NH4+ 0.25
NH4+ 0.30
NH4+ 0.35
NH4+ 0.40
−2 0 2 4 6
VMH
24 better / 0 worse / 0 equal
Glc 0.40
CO2 0.40
Glc 0.35
CO2 0.35
Glc 0.30
CO2 0.30
NH4+ 0.40
Glc 0.25
O2 0.40
CO2 0.25
O2 0.35
NH4+ 0.35
Glc 0.20
NH4+ 0.25
NH4+ 0.30
CO2 0.20
O2 0.30
O2 0.25
Glc 0.15
CO2 0.15
NH4+ 0.20
O2 0.20
NH4+ 0.15
O2 0.15
−2 0 2 4 6
c
E. coli growth (h−1)
CarveMe
7 better / 0 worse / 1 equal
Glc
2-OG
L-Mal
L-Lac
Succ
Glyc
Acetate
Pyr
0.00 0.02 0.04 0.06
BiGG
5 better / 3 worse / 0 equal
L-Lac
L-Mal
Pyr
Acetate
Glyc
Succ
Glc
2-OG
0.00 0.02 0.04 0.06
ModelSEED†
8 better / 0 worse / 0 equal
Glc
Glyc
L-Lac
L-Mal
Pyr
2-OG
Succ
Acetate
0.00 0.02 0.04 0.06
VMH
7 better / 0 worse / 1 equal
Glc
L-Lac
Glyc
Acetate
L-Mal
2-OG
Succ
Pyr
0.00 0.02 0.04 0.06
Absolute-error reduction: better →
Reduction = |Initial − Experiment| − |Post-MQC − Experiment|. Bars sorted within each model.
Hatched: worsened. Zero mark: unchanged. Flux labels include growth rate (h−1). † See source provenance.